

MIVAE: Multiple Imputation based on Variational Auto-Encoder

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ABSTRACT

Nowadays, the issue of MV imputation has become one of the research hotspots in the field of data quality, since the missing values (MVs) are prevalent in real-world datasets and bring challenges to advanced data analytics algorithms. To impute the MVs, most existing approaches directly derive one estimation for each MV, which is categorized as the *single imputation* (SI). However, the SI ignores the uncertainty of the MVs, and thereby usually derive unsatisfactory imputation results compared to the *Multiple imputation* (MI). To extract the uncertainty of the MVs, the MI algorithms derive multiple candidate estimations for each MV. Nevertheless, existing MI approaches are few due to the complicated data-handling process. Accordingly, in this paper, by exploring the Variational Auto-Encoder (VAE) model, we propose a new MI approach, namely MIVAE (Multiple Imputation based on Variational Auto-Encoder) to impute MVs for the tabular data. In MIVAE, we first add a corrupted input layer (where the synthetic MVs are introduced) adjacent to the original input layer to make the model capable of MV issue. Then, we obtain multiple rather than single candidate estimations for each data sample from the posterior distribution of the latent variables learned by our designed model. In such way, the multiple imputation is effectively implemented where the uncertainty of the MVs are extracted perfectly. Next, to obtain satisfactory imputation results, we add a data analysis layer at the end of the network to integrate multiple candidate estimations intelligently. Finally, the experimental results over four real-world datasets demonstrate that MIVAE achieves significantly higher imputation accuracy compared to existing solutions, and MIVAE are capable of handling both numerical and categorized tabular data. For example, the imputation accuracy based on MIVAE improves up to about 40% and 30% compared with PMM and MIWAE (which are the state-of-the-art MI approach) over the CropMapping dataset, respectively. Moreover, we train a MIVAE model over three datasets containing MVs, respectively. By leveraging the trained MIVAE, the classification performance over the imputed data is similar to that over the complete data.

1. Introduction

With the rapid advancement of information technology, data has become an indispensable and fundamental resource. However, due to various uncontrollable factors, e.g., hardware failure, unconscious malfunction, participants refusal, the issue of MVs has appeared widely in real-world datasets, which even induces the problem of data quality (Li and Wang, 2015). The task of *MV imputation*, i.e., replacing the MVs with some plausible ones, is critical because it is a fundamental data preprocessing step for various data analytics tasks, e.g., data mining and knowledge discovery. Over the last decades, various MV imputation algorithms (Ding et al., 2019; Chu et al., 2015) have been proposed, which can be divided into two categories: single imputation (SI) and multiple imputation (MI) (Rubin, 1996). SI uses one single estimation to replace each MV, and most existing MV imputation approaches are SI algorithms. Nevertheless, SI estimates the MVs quantitatively by exploring the correlations amongst data tuples

(or attributes), and ignores the uncertainty of the MVs. In addition, with the explosive growth of data scale and occurrence of complicated missing pattern, the performance of SI is usually unsatisfactory. To alleviate the above deficiency of SI, MI first simulates the posterior distribution of the data, then derives the final imputation result for each MV by fusing multiple estimations sampled from the posterior distribution. However, the posterior distribution of the data (especially the high-dimensional data) is difficult to simulate. Moreover, pervasive repetitive work is required to generate multiple candidate estimations. Therefore, the number of existing MI approaches is limited.

Except for traditional MV imputation algorithms, deep learning, which attracts much attention from academia and industry, provides a new theoretical basis for exploring the inner correlations amongst data. In various deep learning models, VAE is a generative model which provides a probabilistic manner to describe an observation in latent space. Specifically, for each data tuple of the input dataset, the

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encoder of VAE derives an approximate posterior distribution for it in the latent space (i.e., for its latent variables). Then a new data sample is derived by feeding a latent sample (adopted randomly from the approximate posterior distribution) into the *decoder*. VAE shows great potential for multiple imputation in three-folds: (i) The posterior distribution learned by VAE can be leveraged for MI. (ii) VAE is capable of high-dimensional data by enforcing a continuous, smooth latent space representation. (iii) VAE has a complete mathematical theory and is easy to train.

Challenges. Even though there are natural conveniences mentioned above, multiple imputation by directly leveraging the VAE model is challenging in the following aspects:

(1) *The impact of MVs in model training is not considered.* Indeed, a VAE model is usually trained by maximizing the log-likelihood lower bound of the training set without MVs. By contrast, in the model testing phase (i.e., MV imputation phase), the input of the trained model is the data set with MVs whose truths are unknown. In other words, the data distribution of the training set and testing set may not be consistent due to the existence of MVs, which may deteriorate the imputation performance.

(2) *The procedure of multiple imputation is not inherent in VAE.* As introduced earlier, the VAE model only selects a single sample from the approximate posterior distribution of data to recover each data tuple in the original input. Therefore, by leveraging the VAE model directly, the single imputation is achieved rather than multiple imputation. Moreover, the performance of MV imputation by leveraging VAE directly is unsatisfactory (as demonstrated empirically in Section 4.4).

(3) *The fusion mechanism for multiple candidate estimations is imperfect.* Since in the original VAE model, the MI procedure is not inherent, the fusion of multiple candidate estimations is definitely not included. In addition, most existing MI algorithms derive the final imputation results by simply averaging multiple candidate estimations. However, the distance between each candidate estimation and the truth is more likely different in practice, indicating that each candidate estimation should have a specific weight. Therefore, an advanced fusion mechanism for multiple candidate estimations is necessary.

To address the challenges discussed above, we devise a new MV imputation approach named *MIVAE* by exploring VAE model. The major contributions of this paper are summarized as follows.

- We analyze the pitfalls of MV imputation by simply using the VAE model, and accordingly propose a new VAE-based learning model, namely MIVAE, tailored for multiple imputation.
- In the proposed MIVAE, to make the model more capable for MV imputation, we add a corrupted input layer after the original input layer. Specifically, we mark off some values (i.e., synthetic MVs) of the original input and fill the synthetic MVs with some default values (e.g., 0 and mean values). Moreover, to accommodate effective MV imputation, the missing pattern of generated corrupted input in the training set is consistent with that of the test set to be imputed.
- To generate multiple candidate estimations for each MV, we feed multiple latent samples into the *decoder* of the MIVAE, where the samples are randomly selected from the posterior distribution learned by the *encoder*. Additionally, to further improve the imputation accuracy, we add a data analysis layer at the end of the network, which can tackle the difficulty in weighting candidate estimations of each MV.
- We conduct an extensive experimental evaluation using four real-world datasets. The results demonstrate that the proposed MIVAE achieves significantly higher imputation accuracy than existing works over both numerical and categorical data. Moreover, the classification performances over the imputed data derived by MIVAE are closest to those over the complete data. Specifically, the differences of classification accuracy between them over CropMapping, MNIST and Adult are 2%, 0.7%, and 0.4%, respectively.

The remainder of this paper is structured as follows. In Section 2, we introduce the definition of problems, related work and review the VAE model. In Section 3, we describe in detail the MIVAE model and the process of filling missing values based on the MIVAE model. We report the experimental results and finally conclude the paper in Section 5.

2. Preliminaries

In this section, we first introduce some preliminaries relevant to our research. Next, we summarize the related works, and review the VAE model.

2.1. Problem statement

Given n data tuples $X = \{x_1, x_2, \dots, x_n\}$ with d attributes in schema $\mathcal{R} = \{A_1, A_2, \dots, A_d\}$, each data tuple is denoted by $x_i = (x_i[A_1], x_i[A_2], \dots, x_i[A_d]) \in \mathbb{R}^d$, where $x_i[A_j] (1 \leq j \leq d)$ is the observed value of x_i on attribute $A_j \in \mathcal{R}$. A data tuple $x_i \in X$ is a *complete data tuple* if $x_i[A_j]$ is observed for $\forall A_j \in \mathcal{R}$. Otherwise, it is an *incomplete data tuple*. On the other hand, an attribute $A_j \in \mathcal{R}$ is a *complete attribute* if $x_i[A_j]$ is observed for $\forall x_i \in X$. Otherwise it is an *incomplete attribute*. Accordingly, a data set X can be divided into *complete subset* X_{obs} and *incomplete subset* X_{mis} , denoted by $X = (X_{obs}, X_{mis})$, where X_{obs} and X_{mis} consist of all complete data tuples and incomplete data tuples in X , respectively. With the same line, a data tuple $x_i \in X$ can be divided into two parts, denoted by $x_i = (x_i^o, x_i^m)$, where x_i^o and x_i^m consist of values on complete attributes and incomplete attributes, respectively. In addition, to describe the occurrence of MVs in data set X , we introduce a mask matrix $S = (s_1, s_2, \dots, s_n) \in \{0, 1\}^{n \times d}$, where $s_{ij} = 1$ if $x_i[A_j]$ is missing, otherwise $s_{ij} = 0$.

In this work, given a dataset X , the core issue we aim to tackle is to train an effective MIVAE model based on the complete subset X_{obs} and recover the MVs in the incomplete subset X_{mis} based on the learned MIVAE. Moreover, there are three key distinctions warranted to emphasis.

- (1) Our attention focuses on the MV issue itself, i.e., the procedure of MV imputation is independently, not purpose for serving downstream tasks such as clustering and classification.
- (2) We focus on the MV imputation for tabular data including the structural matrix data from relational database and image data. The data with more complicated structures, e.g., graph data and spatial-temporal data, are not included in our research.
- (3) About the missing mechanism, we limit our attention to MAR. Generally, there are three missing mechanisms, i.e., missing completely at random (MCAR), missing at random (MAR), and missing not at random (MNAR). Let $p_\zeta(s | x) = p_\zeta(s | x^o, x^m)$ denote the parametric conditional distribution of the mask s corresponding to a data tuple x , where ζ is the unknown parameter relevant to the missing mechanism. The mathematical forms of the conditional distribution introduced above are as follows.

- MCAR: $p_\zeta(s | x) = p_\zeta(s)$, i.e., the missingness depends on neither the observed values nor the MVs.
- MAR: $p_\zeta(s | x) = p_\zeta(s | x^o)$, i.e., the missingness depends on the observed values rather than the MVs.
- MNAR: $p_\zeta(s | x) = p_\zeta(s | x^o, x^m)$, i.e., the missingness depends on both the observed values and the MVs.

Essentially, MNAR is “non-ignorable” when we deal with the incomplete data set, since the missing mechanism itself has to be modeled. In contrast, both MCAR and MAR are “ignorable” since we do not have to include any information about the missing data itself. The MIVAE proposed in this paper is valid under the assumption of the MAR, which is the most popular mechanism (Schafer and Graham, 2002).

2.2. Related work

In this section, we summarize existing MV imputation works which can be categorized into two classes, i.e., SI (single imputation) and MI (multiple imputation).

2.2.1. Single imputation

As introduced in Introduction, most existing MV imputation methods are SI, which can be further categorized into rule-based imputation, statistical imputation, and machine-learning-based imputation.

The rule-based imputation (Yakout et al., 2011) is usually applied to MVs with predefined relational dependency or editing rules. For example, the method proposed by Song et al. (2020) imputes the MVs in a database by searching its complete neighbors based on editing rules. The statistical imputation characterizes and explores the correlations between the missing values and non-missing values. The typical methods include mean imputation (Evgeniy and Alexei, 2017), k NN imputation (Beretta and Santaniello, 2016; Qiao et al., 2018), hot-deck imputation (Butera et al., 2019), linear regression imputation (Zhang et al., 2019), matrix factorization imputation (Loisel and Takane, 2019; Irawati et al., 2019), etc.

Due to significant research interests, more and more sophisticated deep learning models have been proposed, e.g., autoregressive networks, generative adversarial networks (GANs), rnn networks (Woźniak et al., 2023), convolutional networks (Woźniak et al., 2022), etc. Generally, the rnn networks and convolutional networks are well-suited for processing time series data and image data, respectively. Due to the abilities of unsupervised learning and new data generation, many MV imputation approaches based on autoregressive models and GANs have been proposed. For example, Brett K (Beaulieu-Jones and Moore, 2016) explores the AE (Autoencoders) model for imputing the MVs in the electronic health records (EHRs). MIDA (Gondara and Wang, 2017) imputes the MVs by applying the dAE (denoising AutoEncoder) model. MIDIA (Ma et al., 2020), which is a variant of dAE model, is proposed for MV imputation by considering various missing patterns. GAIN (Yoon et al., 2018), MisGAN (Li et al., 2019), VIGAN (Shang et al., 2017) and GAN-with-GRUI (Luo et al., 2018) generalize the well-known GAN model to MV imputation for both non-sequential and sequential (time series) data.

However, the single imputation pervasively ignores the uncertainty of the MVs, resulting in unsatisfactory imputation accuracy, especially for data with complex structure and distribution. In recent years, multiple imputation is considered as the preferred means to tackle the issue of MVs (Lopes and Ribeiro, 2015).

2.2.2. Multiple imputation

MI is proposed based on Bayes theory. It first replaces each MV with m ($m \geq 1$) candidate estimations to reflect the sampling variability¹. Afterwards, MI combines the m candidate estimations to obtain the final imputation results.

Existing PMM (Olmos and Priyalatha, 2015) derives multiple estimations for each MV by supposing that the error of imputation results or estimated parameters fit for certain distribution. PS (Benedetto et al., 2018) first divides the whole data set into groups by assigning each data tuple a propensity score based on the logistic regression model. Then the MVs in each group are imputed by using Bootstrap inference (Roodman et al., 2019). MCMC (Li et al., 2014) is another typical MI approach, in which the posterior distribution of MVs is derived by establishing a Markov chain. Afterwards, the candidate estimations for each MV are sampled from the posterior distribution. However, the construction of the Markov chain requires massive iterations and is difficult to be applied for high-dimensional data. In addition, the MI approaches introduced above obtain the final imputation results by simply averaging the

multiple candidate estimations. In practice, the distance between each candidate estimation and the truth is more likely different, suggesting that each candidate estimation should have a specific weight.

Except for traditional MI approaches, a few deep-learning-based MI imputation approaches, e.g., MIWAE (Mattei and Frelles, 2018) and not-MIWAE (Ipsen et al., 2021), have been proposed. By maximizing the lower bound of the IWAE (importance-weighted autoencoder) (Burda et al., 2015) model, MIWAE fits a DLVM (Kingma and Welling, 2014) model over the incomplete data set. Afterwards, based on the trained model, MIWAE develops Monte Carlo techniques for single imputation and multiple imputation. Since the number of samples (which is set 10,000 in the original paper) adopted from the posterior distribution is great, the complexity of MV imputation based on MIWAE is high. Compared with MIWAE proposed with MAR assumption, not-MIWAE is proposed for data with MNAR mechanism and suffers the same problem introduced above. MCFlow (Richardson et al., 2020) leverages a normalizing flow model as the underlying density estimator. To address the issue of causality dilemma, MCFlow introduces an iterative learning scheme which alternately updates the underlying density and the imputation results. However, MCFlow often has slow convergence speed due to the iterative learning and may be unstable. Moreover, though MCFlow learns the underlying density of the data, it derives one estimation for each MV, i.e., MCFlow is a single imputation approach essentially.

2.3. Review of VAE model

Next, we briefly review the variational autoencoder (VAE) model to provide some background for our research. As one of the major families of generative models, VAE has shown to be efficient and effective for capturing the latent structure of complex data. Specifically, instead of mapping each input data tuple x_i into a fixed latent vector, VAE maps x_i into a distribution $p_\theta(x_i)$ parameterized by θ . In model training phase, given a training set $\{x_1, x_2, \dots, x_n\}$, VAE aims to maximize the log-likelihood of the training set, as shown in Eq. (1) below,

$$\max_{\theta} \sum_{i=1}^n \log p_\theta(x_i), \quad (1)$$

where $p_\theta(x_i) = \int_{\mathcal{Z}} p_\theta(x_i | z) p(z) dz$, and z is the latent variable encoding by the data sample x_i . Unfortunately, the computation of $p_\theta(x_i)$ is quite difficult due to the involved integrals. Then the $\log p_\theta(x_i)$ is transformed to Eq. (2) by applying variational inference (Blei et al., 2017).

$$\begin{aligned} \log p_\theta(x_i) &= E_{z_i}[\log p_\theta(x_i | z)] \\ &\quad - D_{KL}(q_\phi(z | x_i) \| p(z)) \\ &\quad + D_{KL}(q_\phi(z | x_i) \| p_\theta(z | x_i)) \end{aligned} \quad (2)$$

In the above equation, $p_\theta(x_i | z)$ is the likelihood of x_i by sampling a latent z , and $p(z)$ is the prior distribution of z . Moreover, since $p_\theta(z | x_i)$ is an intractable distribution, a new distribution $q_\phi(z | x_i)$ is introduced to approximate it. However, the KL divergence between $q_\phi(z | x_i)$ and $p_\theta(z | x_i)$, i.e., $D_{KL}(q_\phi(z | x_i) \| p_\theta(z | x_i))$, is still difficult to compute. Fortunately, $D_{KL}(q_\phi(z | x_i) \| p_\theta(z | x_i))$ satisfies the property of $D_{KL}(q_\phi(z | x_i) \| p_\theta(z | x_i)) \geq 0$. Thereby, VAE defines the ELBO (Evidence Lower Bound) of Eq. (2) as the final loss function (denoted by $l(x_i, \theta, \phi)$) by neglecting the KL divergence mentioned above, and as shown in Eq. (3).

$$l(x_i; \theta, \phi) = E_z[\log p_\theta(x_i | z)] - D_{KL}(q_\phi(z | x_i) \| p(z)) \quad (3)$$

Finally, the parameters θ and ϕ are optimized by maximizing the sum of ELBO of the training set (i.e., the loss function), as shown in Eq. (4).

$$\max_{\theta, \phi} \sum_{i=1}^n l(x_i; \theta, \phi) \quad (4)$$

Similar to autoencoder (AE), VAE consists of an *encoder* and a *decoder*. The encoder aims to learn a mapping from input x_i to z (i.e., the distribution $q_\phi(z | x_i)$), and the decoder aims to learn a mapping

¹ sampling variability is the extent to which the measures of a sample differ from the measure of the population

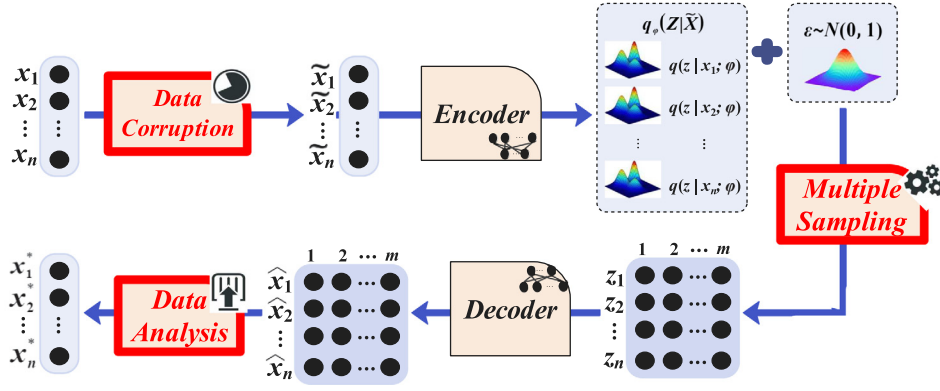


Fig. 1. The network architecture of MIVAE model.

from z back to x_i (i.e., the distribution $p_\theta(x_i | z)$). In summary, VAE provides a probabilistic manner for describing a data tuple in latent space and generates new data samples based on random samples from the latent space. However, VAE assumes that the input data are fully observed to support new data samples generation. If there exist MVs in the input, the recovered input data are similar to the replaced values of MVs rather than the unknown truths. Therefore, some adaptations are required for VAE to tackle MV imputation problem. In this work, we propose a new deep learning model, namely MIVAE, tailored for multiple imputation, carry out a rigid theoretical analysis and perform a holistic research on the problem.

3. MIVAE model

3.1. The architecture of MIVAE model

As introduced earlier, given a data set X , it can be divided into the complete subset X_{obs} and the incomplete subset X_{mis} . Essentially, X_{obs} and X_{mis} can be regarded as two subsets that are independent and identically distributed since the MVs may occur in any data tuple. Therefore, in this work, we use X_{obs} to train an effective MIVAE model which is then used to impute the MVs in X_{mis} . The network architecture of the proposed MIVAE model is shown in Fig. 1. As shown, besides the encoder and decoder, the proposed MIVAE consists of other three main components, i.e., *data corruption*, *multiple sampling* and *data analysis*. The data transformation between each component is described as follows.

(1) *Corrupted input generation*: By introducing synthetic MVs into the training set X_{obs} , a corrupted input $\tilde{X}_{obs}$ is generated through a *data corruption* layer. Specifically, the synthetic MVs are generated by marking off some truths of X_{obs} and filling the marked-off truths with some default values. Generally, zeros or mean values of the corresponding attributes are the most popular choice. For example, the latest MV imputation models E^2 GAN and IWAE adopt zeros, while the MIDIA adopts the mean values or the voting values of the corresponding attributes as default values to replace the synthetically generated MVs. The key idea underlie the corrupted input generation is similar to the denoising auto-encoder which compresses and reconstructs the original input from a destroyed input. However, as literature (Ma et al., 2020) analyzed, the imputation accuracy is suboptimal by directly employing the corruption process of dAE, which is verified again through experiments in Section 4.6. Therefore, in our paper, we adopt the improved algorithm proposed in Ma et al. (2020) to generate the corrupted input layer, where the synthetic MVs are introduced guided by the missing pattern of the testing (incomplete) set and thereby the missing pattern of $\tilde{X}_{obs}$ is consistent with that of X_{mis} .

(2) *Encoding*: For each corrupted data sample $\tilde{x}_i \in \tilde{X}_{obs}$, the *encoder* of MIVAE learns a conditional distribution $q_\phi(z | \tilde{x}_i)$ which is an approximate distribution closed to the intractable posterior $p_\theta(z | x_i)$.

(3) *Multiple sampling*: This component randomly adopts m ($m \geq 1$) latent samples (denoted by $\{z_{i1}, z_{i2}, \dots, z_{im}\}$) from the learned distribution $q_\phi(z | \tilde{x}_i)$.

(4) *Decoding*: The *decoder* of MIVAE outputs m reconstructions of the original input $x_i \in X_{obs}$, denoted by $\{\hat{x}_{i1}, \hat{x}_{i2}, \dots, \hat{x}_{im}\}$.

(5) *Data analysis*: After the decoder, a *data analysis* layer is added to combine multiple reconstructions (candidate estimations) of x_i into a final reconstruction x_i^* . Specifically, in the proposed MIVAE, the data analysis layer consists one or several linear layers, where the activation function is set *None*. For each x_i , this layer weighted rather than simply averaged its m candidate estimations, i.e. $\{\hat{x}_{i1}, \hat{x}_{i2}, \dots, \hat{x}_{im}\}$, linearly. The weight w_{ij} ($1 \leq j \leq m$) corresponding to the candidate estimation $\hat{x}_{ij}$ is learned by the network intelligently.

Compared with the original VAE model, the modifications of the proposed MIVAE are illustrated as follows.

(1) *MIVAE adds a data corruption layer adjacent to the original input layer*. The reason behind this behavior is that in the model testing phase (i.e., MV imputation phase), the input is the incomplete subset X_{mis} where the truths of MVs are unknown. By contrast, in the model training phase, the original input is the complete subset X_{obs} where the impact of MVs is not considered. By introducing synthetic MVs through the data corruption layer, the model is enforced to be more capable of MV imputation.

(2) *Instead of one single latent sample in VAE, MIVAE adopts multiple latent samples from the conditional distribution $q_\phi(z | \tilde{x}_i)$* . The main reasons are two-fold: (i) Multiple candidate estimations are obtained by adopting multiple latent samples, thereby our goal of multiple imputation is achieved. (ii) We can get a tighter log-likelihood lower bound by using multiple samples (Burda et al., 2015). In this way, compared with the VAE, MIVAE achieves a better performance in learning richer latent space representations.

(3) *MIVAE adds a data analysis layer at the end of the network to achieve importance weighted data fusion*. As introduced earlier, most existing MI algorithms obtain the final imputation results by averaging multiple candidate estimations simply. However, the multiple candidate estimations of a missing value should be given different weights. Thus a data analysis layer is added to achieve this goal.

3.2. Theoretical analysis

Given a training set $\{x_1, x_2, \dots, x_n\}$, MIVAE aims to maximize the log-likelihood of $p_\theta(x_i)$ ($1 \leq i \leq n$) with the approximate distribution

$q_\phi(z | \tilde{x}_i)$ rather than $q_\phi(z | x_i)$. The derivation process is shown below.

$$\begin{aligned} \log p_\theta(x_i) &= \log E_{z \sim q_\phi(z|\tilde{x}_i)} \left[\frac{p_\theta(x_i | z) p(z)}{p_\theta(z | x_i)} \cdot \frac{q_\phi(z | \tilde{x}_i)}{q_\phi(z | \tilde{x}_i)} \right] \\ &\geq E_{z \sim q_\phi(z|\tilde{x}_i)} [\log p_\theta(x_i | z) - \log \frac{q_\phi(z | \tilde{x}_i)}{p(z)} \\ &\quad + \log \frac{q_\phi(z | \tilde{x}_i)}{p_\theta(z | x_i)}] \\ &\geq E_{z \sim q_\phi(z|\tilde{x}_i)} [\log p_\theta(x_i | z)] \\ &\quad - D_{KL}(q_\phi(z | \tilde{x}_i) \| p(z)) \\ &\quad + D_{KL}(q_\phi(z | \tilde{x}_i) \| p_\theta(z | x_i)) \end{aligned}$$

By leveraging the property of KL divergence introduced earlier, i.e., $D_{KL}(q_\phi(z | \tilde{x}_i) \| p_\theta(z | x_i)) \geq 0$, we define the lower bound of above equation as

$$\begin{aligned} l_M(x_i; \theta, \phi) &= E_{z \sim q_\phi(z|\tilde{x}_i)} [\log p_\theta(x_i | z)] \\ &\quad - D_{KL}(q_\phi(z | \tilde{x}_i) \| p(z)) \\ &= E_{z_{i1}, z_{i2}, \dots, z_{im} \sim q_\phi(z|\tilde{x}_i)} [\log \frac{1}{m} \sum_{j=1}^m p_\theta(x_i | z_{ij})] \\ &\quad - D_{KL}(q_\phi(z | \tilde{x}_i) \| p(z)). \end{aligned} \quad (5)$$

For item $E_{z \sim q_\phi(z|\tilde{x}_i)} [\log p_\theta(x_i | z)]$, VAE approximates it by selecting a single latent sample z empirically. By contrast, MIVAE approximates it by selecting $m (m > 1)$ latent samples $\{z_{i1}, z_{i2}, \dots, z_{im}\}$, because the lower bound is tighter based on the inference strategy in IWAE (Burda et al., 2015). Moreover, it is notable that the conditional distribution $q_\phi(z | \tilde{x}_i)$ is derived by using the corrupted input $\tilde{x}_i$ rather than the original input x_i . However, the reconstructions based on the sampled latent z are expected to be as close as possible to the original input x_i .

Algorithm 1: MIVAE(X, S, m)

Input : the dataset X with MVs, the mask matrix S , the parameter m for multiple imputation
Output: the imputed dataset X^*

```

1 // dividing  $X$  into the complete subset and the incomplete subset
2  $X = (X_{obs}, X_{mis})$ 
3 // model training phase
4 foreach  $x_i \in X_{obs}$  do
5    $\tilde{x}_i = \iota(x_i)$ 
6    $q_\phi(z | \tilde{x}_i) = \Psi(z | g_\phi(\iota(x_i)))$ 
7   // selecting  $m$  samples from the distribution  $q_\phi(z | \tilde{x}_i)$  randomly
8    $\{z_{i1}, z_{i2}, \dots, z_{im}\} = \text{RandomSample}(q_\phi(z | \tilde{x}_i), m)$ 
9    $p_\theta(x_i | z) = \Phi(x_i | h_\gamma(f_\beta(z_{i1}), f_\beta(z_{i2}), \dots, f_\beta(z_{im})))$ 
10  $\theta^*, \phi^* = \max_{\theta, \phi} \sum_{i=1}^n l_M(x_i; \theta, \phi)$ 
11 // missing value imputation phase
12 // replacing the MVs with some default values
13  $\tilde{X}_{mis} = \text{Replace}(X_{mis}, \text{default\_values})$ 
14  $X_{mis}^* = X_{mis}$ 
15 foreach  $x_j \in \tilde{X}_{mis}$  do
16    $q_{\phi^*}(z | \tilde{x}_j) = \Psi(g_{\phi^*}(x_j))$ 
17    $\{z_{j1}, z_{j2}, \dots, z_{jm}\} = \text{RandomSample}(q_{\phi^*}(z | \tilde{x}_j), m)$ 
18    $x_j^* = h_{\gamma^*}(f_\beta(z_{j1}), f_\beta(z_{j2}), \dots, f_\beta(z_{jm}))$ 
19    $X_{mis}^*[j] = x_j^*$ 
20    $X^* = (X_{obs}, (1 - S)X_{mis} + SX_{mis}^*)$ 
21 return  $X^*$ 
```

The specific formulation of each item in Eq. (5) is illustrated below.

$$p_\theta(x_i | z) = \Phi(x_i | h_\gamma(f_\beta(z_{i1}), f_\beta(z_{i2}), \dots, f_\beta(z_{im}))),$$

where

- $(\Phi(\cdot | h))_{h \in H}$: is a parametric families of distributions. Generally, Φ is assumed to be Gaussian distribution/multinomial distribution for continuous data/categorical data.

- f_β : is the function of the *decoder*, where the parameter β is learned through the deep neural networks. Moreover, various applicable networks can be applied to various data types. For example, we can adopt the fully-connected networks for numerical data, convolutional networks for image data, and recurrent networks for time series. By employing f_β , multiple candidates are derived for each original input x_i .
- h_γ : is a function used for combining the multiple candidates derived by the decoder. The parameter γ is learned through the *data analysis layer* which is a linear weighted regression function.

$$q_\phi(z | \tilde{x}_i) = \Psi(z | g_\phi(\iota(x_i))),$$

where

- $(\Psi(\cdot | g))_{g \in K}$: is also a parametric family of simple distributions, which is assumed to be normal distribution in our work.
- g_ϕ : is the function of the *encoder* where the parameter ϕ is also learned through the deep neural networks like the *decoder*.
- ι : is a function of data corruption used to introduce synthetic MVs into the original input x_i and generate the corrupted input $\tilde{x}_i$, i.e., $\tilde{x}_i = \iota(x_i)$.

Finally, the *loss function* of the proposed MIVAE is shown in Eq. (6).

$$\max_{\theta, \phi} \sum_{i=1}^n l_M(x_i; \theta, \phi) \quad (6)$$

VAE essentially adds ‘‘Gaussian noise’’ to encoder results and dynamically adjusts the noise intensity so that the results obtained by decoder can be robust to noise. In contrast, in addition to the regular Gaussian noise, MIVAE introduces noise caused by the presence of missing values (i.e., the error between the default substitute value of the missing value and the ground truth). By minimizing the sample’s reconstruction error (the error between the reduced value and the true value), the model is more robust, i.e., the output obtained based on damaged input can restore the real input as much as possible, so as to be more conducive to filling the missing value.

3.3. MV imputation based on MIVAE

Given a data set $X = (X_{obs}, X_{mis})$, we aim to impute the MVs in the incomplete subset X_{mis} by employing the MIVAE model trained based on the complete subset X_{obs} . The pseudo-code is shown in Algorithm 1. Specifically, the model training follows the steps of data transformation introduced in Section 3.1. It is worth noting that the missing pattern of the corrupted input $\tilde{X}_{obs}$ is consistent with that of the incomplete subset X_{mis} to be imputed. For missing pattern determination and the corrupted input generation, we adopt existing strategies proposed in Ma et al. (2020).

Example 1. Table 1 shows a data fragment of Bias² dataset which consists of summer data from 2013 to 2017 used for bias correction of next-day maximum and minimum air temperatures. Each data tuple (shown as a row in Table 1) has 23 attributes, and only a few of them are shown in Table 1 due to the space limitation. For illustration, we use ‘?’ to indicate the MVs. The model training process of MIVAE based on the complete subset X_{obs} in Table 1 is shown in Fig. 2. Note that for the corrupted input generation in the example, we replace the synthetic MVs with the mean values of corresponding attributes. For example, we replace the synthetic MVs on the first, second and the last attributes of the data with the mean values 0.4695, 0.7169 and 0.6097, respectively.

² <https://archive.ics.uci.edu/ml/datasets/Bias+correction+of+numerical+prediction+model+temperature+forecast>.

Table 1
Bias data fragment.

	Tmax	Tmin	WS	RHmax	...	RHmin	Slope
x_1	0.3948	0.7614	0.3186	0.2115	...	0.4796	0.1652
x_2	0.1574	0.5899	0.7221	0.5538	...	0.0974	0.5107
x_3	0.6275	0.7637	0.5056	0.5910	...	0.9117	0.8044
x_4	0.3773	0.8114	0.5831	?	...	?	0.9207
x_5	?	0.6133	0.6062	0.6176	...	0.1490	0.6096
x_6	0.4821	0.6311	0.7804	0.9316	...	0.0068	0.5732
x_7	0.8594	0.8902	0.4078	0.6032	...	0.1178	0.9243
x_8	0.3237	0.8485	0.7373	0.6276	...	0.5095	0.9054
x_9	?	0.9865	0.6809	?	...	?	0.6079
x_{10}	0.3761	0.6505	0.7414	0.7055	...	0.5289	0.6199
x_{11}	0.1595	0.2932	0.7435	0.6489	...	0.1490	0.3705
x_{12}	0.5896	0.8366	0.6133	0.6426	...	1.0000	0.6773
x_{13}	0.3773	0.8114	0.5831	?	...	?	0.9207
x_{14}	?	0.9865	0.6809	?	...	?	0.6079
x_{15}	?	?	0.4711	0.6865	...	0.1033	?
x_{16}	0.6404	?	0.5578	0.5140	...	?	0.8637
x_{17}	0.4206	0.6149	0.0668	0.7483	...	0.034	0.6930
...	...	...	...	...	...	...	...

In the MV imputation phase, we first replace the MVs in the incomplete subset X_{mis} with the user-specified default values and then take the initialized X_{mis} as the corrupted input. Afterwards, via the trained MIVAE model, the MVs are reconstructed through the encoding, decoding, and data analyzing steps. Note that the output of the MIVAE model is the reconstructions of the whole data set, while we only need the reconstructions of the MVs. Therefore, the final imputed data set X^* is

$$X^* = (X_{obs} \cdot (1 - S) + S X_{mis}^*). \quad (7)$$

Example 2. Based on the MIVAE model trained in Example 1, the MV imputation for the incomplete subset X_{mis} in Table 1 is shown in Fig. 3.

4. Experiments

4.1. Experimental settings

Datasets: To evaluate the performance of the proposed MIVAE, we adopt four real-world datasets.

- **Bias:** As introduced earlier, the data is related to the air temperatures which are predicted based on the LDAPS model operated by the Korea Meteorological Administration. It consists of 7750 data tuples where each data tuple has 23 attributes.
- **CropMapping³:** consists of fused bi-temporal optical-radar data for cropland classification. Each data tuple has 2*49 radar features and 2*38 optical features from two dates, i.e., 05 and 14 July 2012. In the original dataset, there are 325,834 data tuples, which is a big data set. To reduce the time costs of various approaches, we conduct the experiments over the 30,000 data tuples sampled randomly from the entire dataset.
- **MNIST⁴:** is a large database of handwritten digits widely used in the field of machine learning. It contains 70,000 images totally, each of which is originally organized as a pixel matrix with size 28*28. Since the input of the proposed model is tabular data, we transform the data matrix of each image to a one-dimensional vector with 784 attributes.
- **Adult⁵:** is a dataset of census information, which consists of 32,000 observations with 12 attributes including age, workclass, fnlwgt, education, education-num, marital-status, occupation, relationship, race, sex, hours-per-week and native-country.

³ <http://archive.ics.uci.edu/ml/datasets/Crop+mapping+using+fused+optical-radar+data+set>.

⁴ <http://yann.lecun.com/exdb/mnist/>.

⁵ <http://archive.ics.uci.edu/ml/datasets/Adult>.

In the first three datasets, i.e., Bias, CropMapping and MNIST, all attributes are numerical, while in the Adult dataset, there are 4 numerical attributes and 8 categorical attributes. We use the first three datasets and last dataset to evaluate the performance of various imputation methods over the numerical data and categorical data, respectively. In Bias and Adult, the MVs exist naturally, but the corresponding truths are unfortunately unknown. To measure the imputation accuracy, we delete the incomplete data tuples (where the ratio is less than 5%) in the original Bias and Adult. In CropMapping and MNIST, there is no MV. Thereby, we consider all the above four datasets are initially clean, and generate the incomplete datasets with various missing ratios manually. To generate the MVs fitting MAR mechanism, a subset of attributes are first randomly selected to be fully-observed. Then, only the remaining attributes (referred to *candidate incomplete attributes*) may have MVs. Specifically, for Bias, CropMapping and Adult, we split the whole dataset into *training set* and *testing set* based on the missing ratio. For example, suppose the missing ratio is 5%. In that case, we select 5% data tuples from the dataset randomly as the incomplete subset (testing set) to report the imputation results, while the remaining data tuples constitute the complete subset (training set) to train the learning models. In addition, for each incomplete data tuple, we select a half number of attributes from the candidate incomplete attributes randomly and mark off the values on the selected attributes to simulate missingness. For MNIST, since there are inherent training set and testing set, the number of values marked off is determined based on the missing ratio for each (incomplete) data tuple in testing set. For example, suppose the missing ratio is 5%. We select 5% attributes randomly from the candidate incomplete attributes for each (incomplete) data tuple in testing set and mark off the values on the selected attributes.

Algorithms for comparison: We compare the proposed MIVAE with 12 representative existing MV imputation algorithms which are categorized into two groups.

The single imputation approaches include:

- **Mean** (Evgeniy and Alexei, 2017) replaces the MVs on each incomplete attribute with the mean value of the corresponding attribute.
- **KNN** (Beretta and Santaniello, 2016) estimates the MVs in an incomplete data tuple using the distance-weighted average over its k neighbors.
- **MLR** (Zhang et al., 2019) imputes the MVs based on a linear regression model where the linear correlations between the incomplete attributes and complete attributes are explored.
- **MissForest** (Stekhoven and Bühlmann, 2011) is a non-parametric imputation method that requires the data tuple to be pairwise independent.
- **SVM** (Pal and Foody, 2010) is a supervised learning method for regression and classification. We build a SVM model to extract the correlations between the incomplete attributes and complete attributes for MV imputation.
- **DT** (Beaulac and Rosenthal, 2020) is another non-parametric supervised learning method used for regression.
- **MIDIA** (Ma et al., 2020) is a variant of dAE model tailored for MV imputation. It is effective for MVs with various missing patterns.
- **SIVAE** is the algorithm we implemented by directly using the original VAE model for comparison.
- **GAIN** (Yoon et al., 2018) imputes the MVs conditioned on the observed component of input data by exploring the well-known generative adversarial network (GAN).
- **MCFlow** (Richardson et al., 2020) utilizes the normalizing flow generative models to learn exact density estimation on the incomplete data via an iterative learning schema.

The multiple imputation approaches include:

- **MICE** (Buuren and Groothuis-Oudshoorn, 2011) is a principled multiple imputation method used to replace the MVs based on a regression model which is trained iteratively.

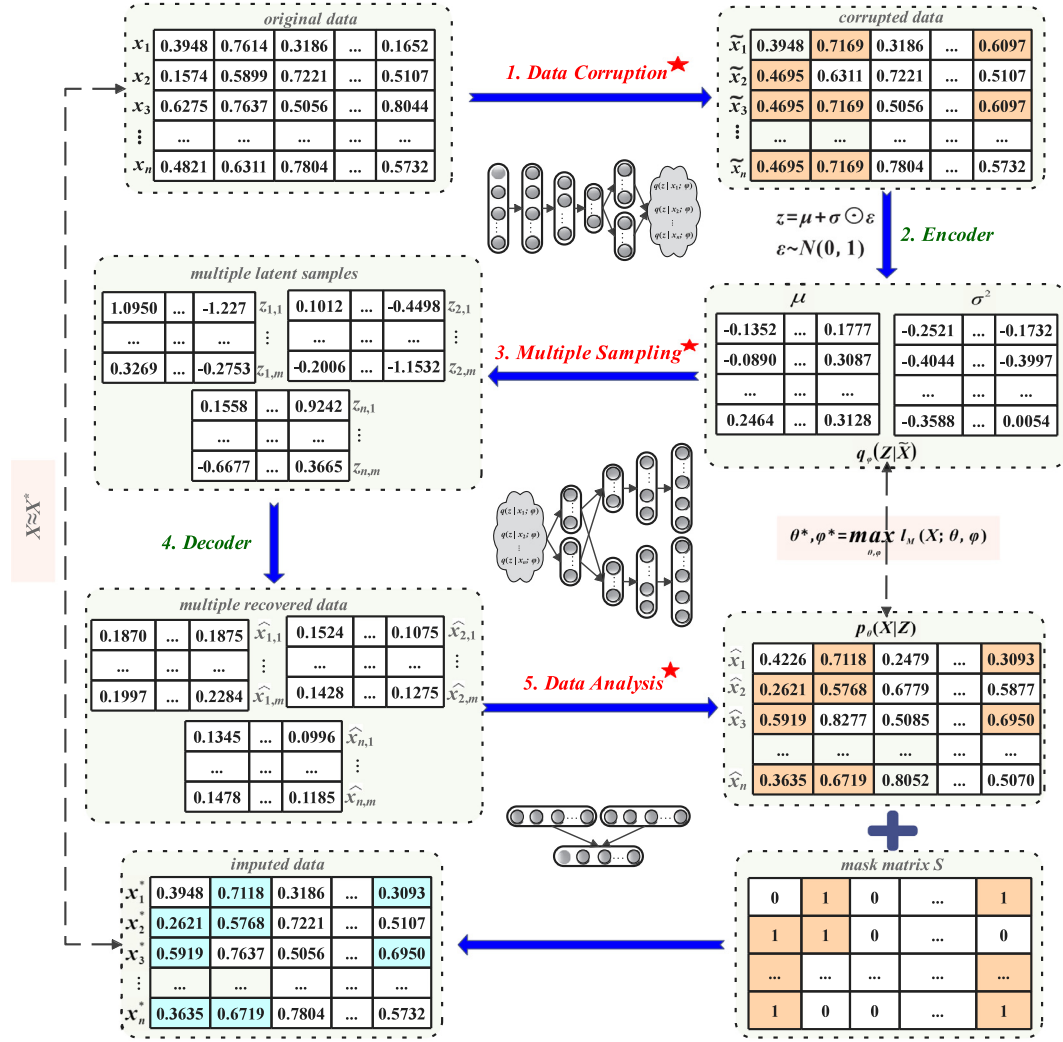


Fig. 2. The model training process of MIVAE.

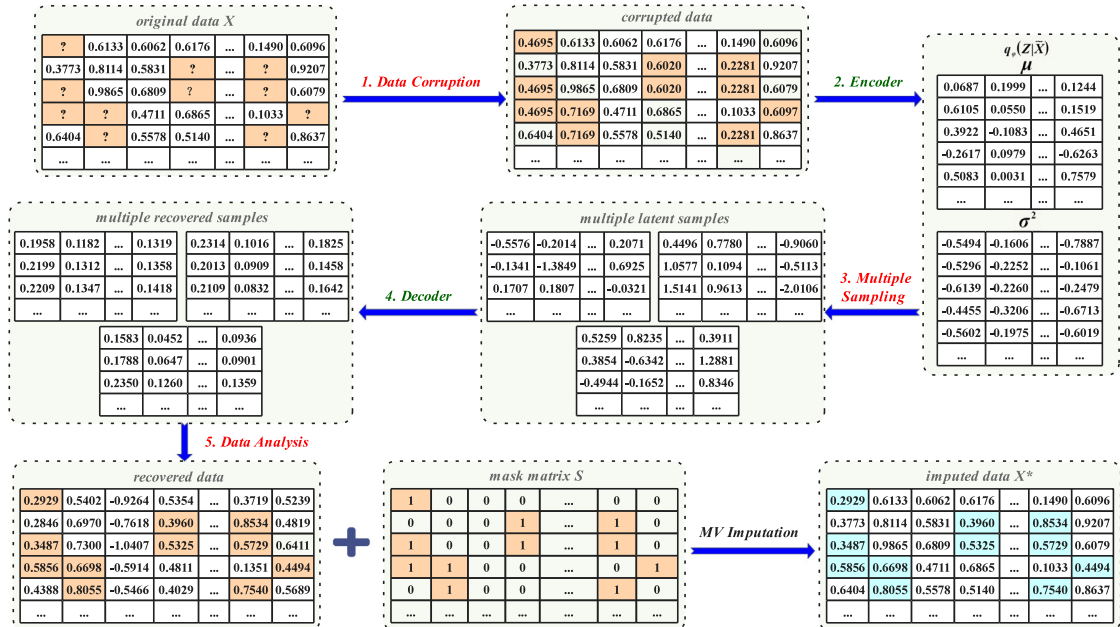


Fig. 3. The MV imputation process of MIVAE.

Table 2
The assumptions and applicable conditions of various algorithms.

Algorithm	Missing pattern	Required data	Data types
Mean	MCAR	Incomplete	Continuous
KNN	MCAR	Incomplete	Mixed
MLR/PMM	MCAR	Fully-Observed	Continuous
MissForest/DT/SVM/HyperImpute	MAR	Incomplete	Mixed
MIDA	MAR	Fully-Observed	Mixed
MIDIA	MAR	Fully-Observed	Mixed
GAIN/MCFlow/misGAN/Ganmin/HIVAE	MCAR	Incomplete	Mixed
MICE	MAR	Incomplete	Mixed
MIWAE	MAR	Incomplete	Mixed
MIVAE/SIVAE	MAR	Fully-Observed	Mixed

Table 3
Architecture details and hyperparameters of MIVAE over four datasets.

Dataset	Layer numbers	Training epochs	Learning rate	Batch_size	Optimizer
Bias	FC(15, 13)-FC(13-11)-FC(11-15)	200	0.01	256	Adam
CropMapping	FC(174, 124)-FC(124-124)-FC(124-174)	200	0.001	256	Adam
MNIST	FC(784, 512)-FC(512-128)-FC(128-128)	200	0.001	256	Adam
Adult	FC(102, 51)-FC(51-51)-FC(51-51)	100	0.001	256	Adam

- *PMM* (Olmos and Priyalatha, 2015) generates multiple candidate estimations for each MV by employing the linear regression model with the assumption that the parameters of the imputation model fit for certain distribution.
- *MIDA* (Gondara and Wang, 2017) uses overcomplete DAE as the base model and implements multiple imputation by initializing the model with a different set of random weights at each run.
- *MIWAE* (Mattei and Frellsen, 2018) implements the multiple imputation based on the importance-weighted autoencoder.

Table 2 summarizes the assumptions and applicable conditions of various baselines and our MIVAE. As stated in the Introduction, since the missingness depends on the missing components themselves, it is impossible to learn the missing data distribution without imposing domain-specific assumptions, constraints for the missingness mechanism (Ipsen et al., 2021). Thus, we restrict our discussion to the (most commonly studied) MCAR and MAR settings. Moreover, note that the missing pattern of various approaches listed in the table denotes the most general regime under which they are appropriate. For MLR, PMM, MIDA, MIDIA, SIVAE and the proposed MIVAE, they first train the imputation models using fully-observed training set, and then employs the trained model to impute the MVs in the incomplete testing set. For the remaining approaches, the processes of model training and MV imputation are integrated, especially for the iterative methods. Finally, most approaches are capable of handling different data types except for mean imputation and regression models (i.e., MLR and PMM).

Model Settings: As mentioned earlier, the most popular default values for replacing the MVs in the training set are zeros or the mean values of the corresponding attributes. Moreover, through the experimental evaluation (in Section 4.3), we find that the imputation accuracies of MIVAE with default values as zeros and mean/voting values are similar. Therefore, in this paper, for various missing ratios, the synthetically introduced MVs in Bias and CropMapping are replaced with the mean values for the numerical data, while the synthetically introduced MVs in Adult are replaced with the voting values for the categorical data. Moreover, the pixel point in the image of MNIST is black when the pixel value is zero, which inherently reflects the data missing. Thus, we use zeros to replace the synthetical MVs for MNIST. To obtain reliable experimental results, we report the average results of 10 times repeats for each test. For KNN, the parameter k is set as 10 since low imputation errors within an acceptable time cost for the four datasets are reached with $k = 10$. We implement MissForest, DT, SVM, and MICE by using the *sklearn* package in Python. We adopt the original network architectures in our experiments for the deep-learning-based approaches, e.g., MIDA, MIDIA, GAIN, and MIWAE. For SIVAE, we use the same network architecture as the proposed MIVAE model. The

specific network architecture and hyper-parameters of the proposed model are shown in Table 3. Since network architectures of encoder and decoder are symmetrical in MIVAE, we only list the number of units of each layer in encoder. For all datasets, there are four layers in encoder, where the first layer is the input layer and the last layer is the middle hidden layer.

Measurements: We adopt the following measurements to evaluate the performance of MIVAE.

- **RMSE:** For numerical data, we adopt the *Root Mean Square Error* (RMSE) as the measure to evaluate the imputation accuracy. If the imputation results are closer to the ground truths, the RMSE is lower and thereby the imputation performs better.
- **Macro-F:** For categorical data, we measure the performance by using the *Macro-F* (Sokolova and Lapalme, 2009) which is the averaging of each possible value for the given data. The higher Macro-F is, the performance of MV imputation is better.
- **Accuracy and Kappa:** We adopt the *accuracy* and *kappa* (Pontius and Millones, 2011) to evaluate the performance of classification. A higher accuracy (or kappa) indicates the better imputation performance.

4.2. Selection of parameter m

Recall that the parameter m is the number of candidate estimations obtained for each MV, which impacts the performance of the proposed MIVAE. Thus we evaluate the imputation accuracy of MIVAE by varying m over the four real-world datasets with the missing ratio of 30%. Theoretically, with a larger m , the uncertainty of the MVs is more likely to be captured and the lower bound of the model is tighter. On the other hand, the time costs of model training are higher because more parameters need to be learned. As shown in Fig. 4, the RMSE (Micro-F) of MIVAE decreases (increases) as m increases over Bias, CropMapping and MNIST (Adult), which is consistent with our theoretical analysis. To trade-off the effectiveness and efficiency, we adopt $m = 20$ as default.

4.3. Selection of default values

In this section, we verify the impact of default values during corrupted input generation to the final imputation accuracy of the proposed MIVAE. For the numerical data, i.e., Bias, CropMapping and MNIST, the candidate default values are zeros and the mean values of the corresponding attributes. For the categorical data, i.e., Adult, the candidate default values are zeros and the voting values of the corresponding attributes. As shown in Fig. 5, the imputation accuracies of MIVAE with different default values are similar to each other. For the

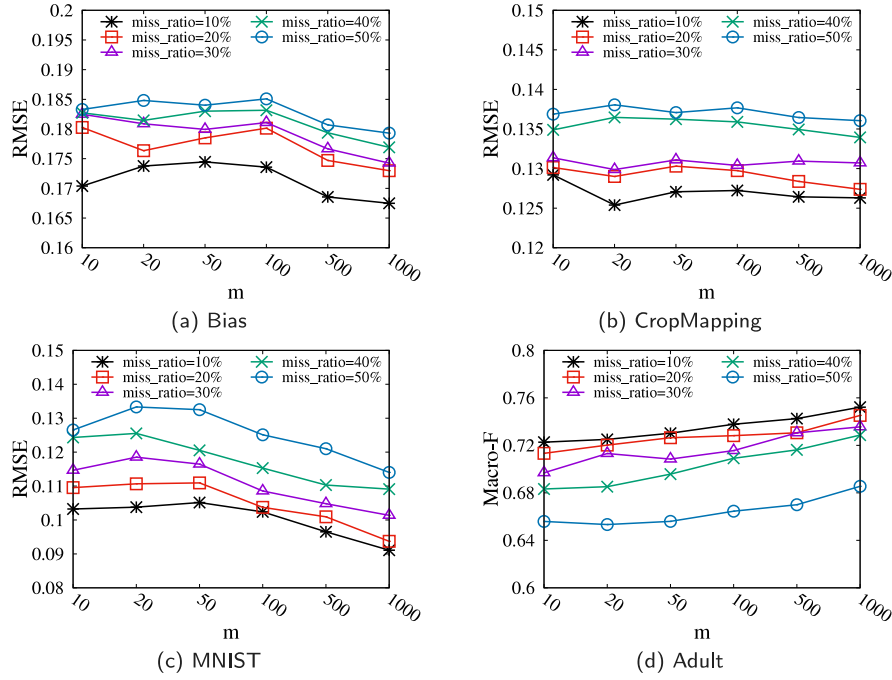
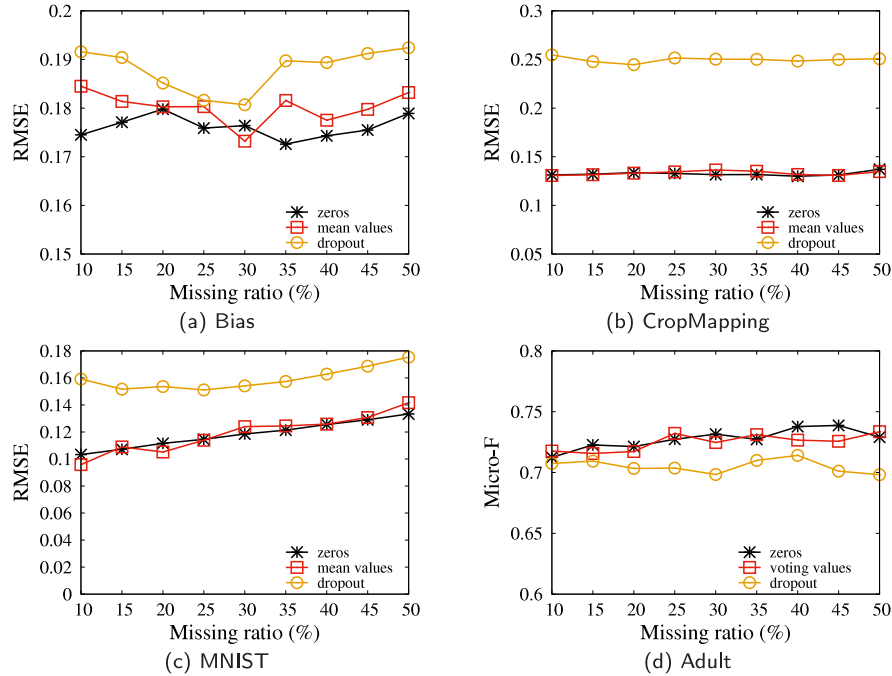
Fig. 4. The selection of parameter m .

Fig. 5. The selection of default values.

four datasets we adopted, with various missing ratios, the maximum differences between different default values are 0.01, 0.002, 0.008 and 0.013, respectively.

In addition, to verify the effectiveness of the data corruption process adopted in this paper, we show the imputation accuracies of MIVAE by directly employing the data corruption process of dAE (denoted by dropout) in Fig. 5. As shown, the imputation accuracies of MIVAE with dropout is obviously worse than those of MIVAE with zeros or mean/voting values, which verifies again that the data corruption process adopted in MIVAE is more effective than directly employing the strategy proposed in dAE.

4.4. Evaluation on imputation accuracy

In this section, we compare the imputation accuracy of our proposed MIVAE with existing methods. As shown in Tables 4–7, the proposed MIVAE achieves the best performance in most cases. Since SIVAE achieves the single imputation by directly employing the existing VAE model which is not designed for MV imputation, its performance is even worse than the simple Mean imputation in some cases. This also reflects the effectiveness of the modifications in our MIVAE model. Since the impact of MVs is not considered in the model training phase, the performance of MIWAE is unsatisfactory even though it develops a tighter lower bound compared with the VAE model. For PMM, it

Table 4

The imputation accuracy over Bias.

	10%	15%	20%	25%	30%	35%	40%	45%	50%
Mean	0.2387	0.2369	0.2370	0.2385	0.2397	0.2363	0.2363	0.2367	0.2377
KNN	0.2187	0.2182	0.2184	0.2183	0.2189	0.2188	0.2208	0.2177	0.2198
MLR	0.2208	0.2170	0.2168	0.2173	0.2188	0.2176	0.2184	0.2167	0.2194
MissForest	0.2228	0.2225	0.2254	0.2118	0.2156	0.2148	0.2160	0.2151	0.2549
DT	0.2961	0.2997	0.2953	0.2941	0.2984	0.2940	0.2987	0.2985	0.2984
SVM	0.2174	0.2117	0.2144	0.2145	0.2156	0.2158	0.2165	0.2141	0.2159
MIDIA	0.1865	0.1867	0.1853	0.1803	0.1871	0.1903	0.1874	0.1823	0.1863
SIVAE	0.2409	0.2251	0.2181	0.2518	0.2077	0.2253	0.2167	0.2155	0.2087
GAIN	0.2113	0.2126	0.2057	0.2107	0.2261	0.2021	0.2033	0.2314	0.2077
MCFlow	0.1860	0.1914	0.1897	0.1828	0.1910	0.1868	0.1897	0.1925	0.1971
MICE	0.1899	0.1895	0.1909	0.1858	0.1911	0.1884	0.1890	0.1888	0.1843
PMM	0.2193	0.2225	0.2184	0.2146	0.2185	0.2179	0.2192	0.2175	0.2189
MIDA	0.2017	0.1899	0.1881	0.1880	0.1966	0.1888	0.1955	0.1885	0.1907
MIWAE	0.1865	0.1885	0.1845	0.1814	0.1858	0.1872	0.1910	0.1904	0.1927
MIVAE	0.1745	0.1771	0.1798	0.1759	0.1764	0.1726	0.1743	0.1755	0.1789
<i>Improvement</i>	6.58%	5.42%	2.63%	2.45%	5.36%	8.21%	7.52%	3.89%	3.00%

Table 5

The imputation accuracy over CropMapping.

	10%	15%	20%	25%	30%	35%	40%	45%	50%
Mean	0.2542	0.2467	0.2440	0.2500	0.2499	0.2491	0.2469	0.2490	0.2501
KNN	0.1858	0.1799	0.1756	0.1839	0.1829	0.1861	0.1844	0.1852	0.1887
MLR	0.1922	0.1783	0.1805	0.1914	0.1839	0.1943	0.1852	0.1946	0.1777
MissForest	0.2731	0.2754	0.2685	0.2736	0.2721	0.2762	0.2731	0.2782	0.2779
DT	0.1522	0.1478	0.1501	0.1487	0.1486	0.1456	0.1498	0.1500	0.1497
SVM	0.1443	0.1405	0.1418	0.1450	0.1440	0.1471	0.1441	0.1463	0.1453
MIDIA	0.1364	0.1331	0.1318	0.1446	0.1339	0.1424	0.1383	0.1405	0.1502
SIVAE	0.3743	0.2479	0.4842	0.2529	0.2517	0.2527	0.4152	0.2549	0.2526
GAIN	0.1498	0.1548	0.1528	0.1514	0.1508	0.1564	0.1520	0.1479	0.1532
MCFlow	0.1781	0.1600	0.1728	0.1843	0.1691	0.1586	0.1616	0.1725	0.2052
MICE	0.1403	0.1354	0.1368	0.1368	0.1366	0.1406	0.1377	0.1384	0.1363
PMM	0.1722	0.1683	0.1650	0.1737	0.1740	0.1817	0.1733	0.1682	0.1655
MIDA	0.1371	0.1365	0.1370	0.1400	0.1382	0.1424	0.1402	0.1400	0.1539
MIWAE	0.2522	0.2402	0.2386	0.2460	0.2485	0.2453	0.2436	0.2472	0.2468
MIVAE	0.1313	0.1320	0.1336	0.1329	0.1316	0.1317	0.1300	0.1313	0.1371
<i>Improvement</i>	3.92%	0.82%	-1.34%	2.95%	1.76%	6.75%	5.98%	5.41%	-0.60%

Table 6

The imputation accuracy over MNIST.

	10%	15%	20%	25%	30%	35%	40%	45%	50%
Mean	0.3599	0.3601	0.3609	0.3602	0.3604	0.3603	0.3604	0.3602	0.3603
KNN	0.1727	0.1733	0.1738	0.1743	0.1751	0.1753	0.1762	0.1773	0.1779
MLR	0.3285	0.3317	0.3352	0.3373	0.3400	0.3423	0.3442	0.3468	0.3486
MissForest	0.2235	0.2381	0.2588	0.2687	0.2843	0.2933	0.3238	0.3341	0.3564
DT	0.3938	0.3942	0.3952	0.3945	0.3947	0.3946	0.3947	0.3945	0.3945
SVM	0.2902	0.2903	0.2909	0.2905	0.2908	0.2905	0.2907	0.2909	0.2906
MIDIA	0.1374	0.1295	0.1329	0.1323	0.1390	0.1415	0.1455	0.1518	0.1580
SIVAE	0.3484	0.3489	0.3510	0.3511	0.3531	0.3546	0.3551	0.3566	0.3585
GAIN	0.1308	0.1317	0.1464	0.1632	0.1750	0.1816	0.2016	0.2060	0.2071
MCFlow	0.1171	0.1214	0.1181	0.1226	0.1261	0.1303	0.1362	0.1428	0.1491
MICE	0.2398	0.2472	0.2597	0.2968	0.2982	0.3015	0.3207	0.3465	0.3360
PMM	0.3236	0.3304	0.3306	0.3279	0.3306	0.3286	0.3246	0.3287	0.3438
MIDA	0.1289	0.1300	0.1350	0.1405	0.1467	0.1536	0.1616	0.1712	0.1793
MIWAE	0.2726	0.2797	0.3182	0.2826	0.2878	0.2838	0.2917	0.3094	0.2961
MIVAE	0.1032	0.1072	0.1116	0.1145	0.1185	0.1213	0.1255	0.1290	0.1333
<i>Improvement</i>	13.36%	13.23%	5.82%	7.00%	6.39%	7.35%	8.51%	10.67%	11.81%

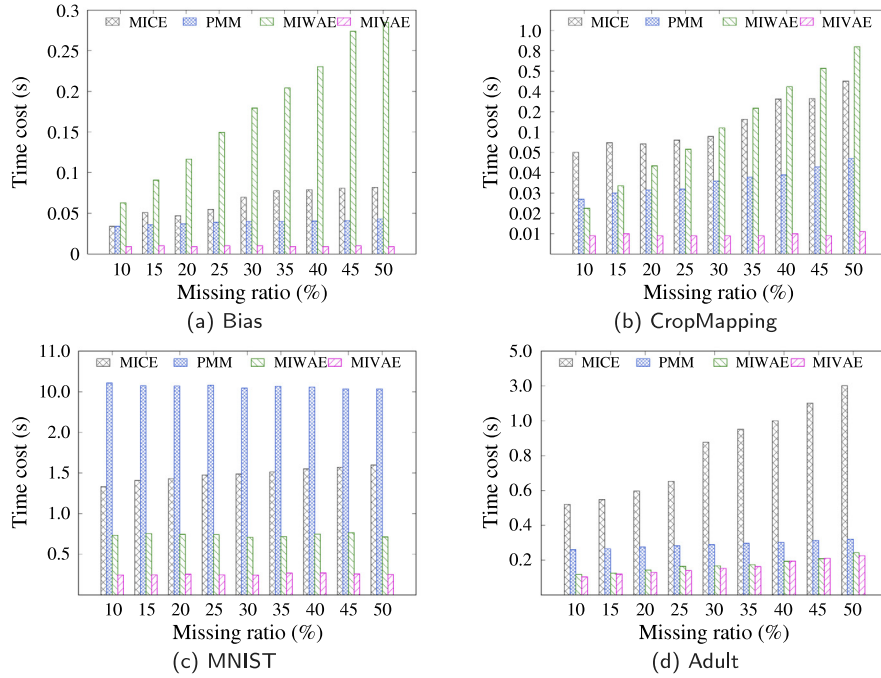
achieves the multiple imputation based on the linear regression model which is deficient to capture the complex correlations of the data. Thus its performance is better than MLR but worse than most of the advanced SI methods, e.g., SVM, MIDA, and MIDIA. It is worth noting that the performance of DT is well-behaved over CropMapping and Adult but ill-behaved over Bias and MNIST. One of the possible reasons is that the attribute divergence of Bias (MNIST) is unclear, which results in the step of feature selection being difficult for DT. Further, it also illustrates that no one approach is capable of all datasets absolutely. For MIDA, MIDIA, GAIN and MCFlow which leverage advanced machine

learning techniques, their performances are better than the traditional approaches in most cases. One possible reason is that they are meticulously designed to tackle the problem of MV imputation. Another possible reason is that the MVs are introduced as “noises” during model training, which makes the learned model robust and capable of MV imputation. Since GAIN is designed for numerical data, it has a competitive performance over the numerical datasets. As MCFlow preserves and recovers the semantic content of the intended image, it does a good job on the MNIST. Finally, by drawing lessons from MIDIA, exploring the rationale of MI and VAE, the proposed MIVAE

Table 7

The imputation accuracy over Adult.

	10%	15%	20%	25%	30%	35%	40%	45%	50%
Mean	0.5332	0.5237	0.5382	0.5353	0.5336	0.5352	0.5378	0.5366	0.5383
KNN	0.7298	0.7065	0.7123	0.7178	0.7071	0.6976	0.6910	0.6789	0.6778
MLR	0.3856	0.4015	0.3897	0.3803	0.3737	0.3739	0.3516	0.3482	0.3348
MissForest	0.3123	0.3116	0.3608	0.3267	0.3244	0.2953	0.3621	0.2038	0.3235
DT	0.7298	0.7147	0.7171	0.7107	0.7190	0.6702	0.6511	0.6205	0.5785
SVM	0.3391	0.3606	0.3270	0.3082	0.3274	0.3069	0.3111	0.2874	0.3144
MIDA	0.6676	0.6959	0.7047	0.7002	0.7187	0.7073	0.7123	0.7122	0.7007
SIVAE	0.5266	0.5336	0.5677	0.5492	0.5283	0.5283	0.5625	0.5334	0.5376
GAIN	0.4049	0.4049	0.3280	0.3317	0.3712	0.3279	0.3239	0.3564	0.3700
MCFlow	0.6787	0.6443	0.6257	0.6176	0.5531	0.4741	0.5227	0.5342	0.5040
MICE	0.7042	0.7082	0.7000	0.7085	0.6949	0.6794	0.6484	0.6411	0.6156
PMM	0.4574	0.4604	0.4433	0.4536	0.4344	0.4529	0.4149	0.4532	0.4191
MIDA	0.6578	0.6887	0.7100	0.7119	0.7045	0.7055	0.7025	0.6927	0.6746
MIWAE	0.5160	0.5362	0.5420	0.5344	0.5408	0.5345	0.5403	0.5382	0.5350
MIVAE	0.7124	0.7226	0.7213	0.7273	0.7315	0.7272	0.7377	0.7387	0.7291
Improvement	-2.00%	0.44%	1.58%	1.37%	1.78%	2.81%	2.49%	2.23%	4.98%

**Fig. 6.** The comparison of time costs.

is effectively designed and achieves the best performance with various missing ratios over the four real-world datasets. To show the effectiveness of MIVAE clearer, we list the imputation accuracy improvement of MIVAE compared to the next-best-method in the last line of [Tables 4–7](#). For the three numerical datasets, i.e., Bias, CropMapping and MNIST, MIVAE produces an optimal reductions in RMSE of 8%, 7% and 13%, respectively. For the categorical dataset, i.e., Adult, MIVAE produces an optimal increase in Macro-F of 5%. Moreover, the experimental results verify that the proposed MIVAE supports both numerical and categorical data.

4.5. Evaluation on imputation efficiency

Besides the superiority of MIVAE in imputation accuracy, in this section, we further evaluate the performance of MIVAE in imputation efficiency. Since once an imputation model is trained, it is used for MV imputation repeatedly, thereby we only consider the time costs

of the model testing (i.e., the MV imputation process). On the other hand, due to the complicated process of generation and fusion for multiple candidate estimations, in most cases, the time costs of MI approaches are greater than those of SI approaches inevitably. Thus we just compare the time costs of MIVAE with existing MI approaches (i.e., MICE, PMM and MIWAE). As shown in [Fig. 6](#), the time costs rise with the increase of missing ratio for all MI approaches over the Bias, CropMapping, and Adult. Since for these three datasets, the missing ratio is higher, there is a larger testing set as introduced in [Section 4.1](#). By contrast, the MNIST dataset has a testing set with fixed size, thus the time costs remain stable with the increase of missing ratio for MNIST. Moreover, the time cost of the proposed MIVAE is the lowest over all datasets, which verifies that MIVAE has the best performance in imputation efficiency compared with the existing MI approaches. In addition, another interesting phenomenon is observed in [Fig. 6](#). The time cost of PMM is lower than those of MICE and MIWAE over the Bias, CropMapping and Adult, but significantly higher than those of

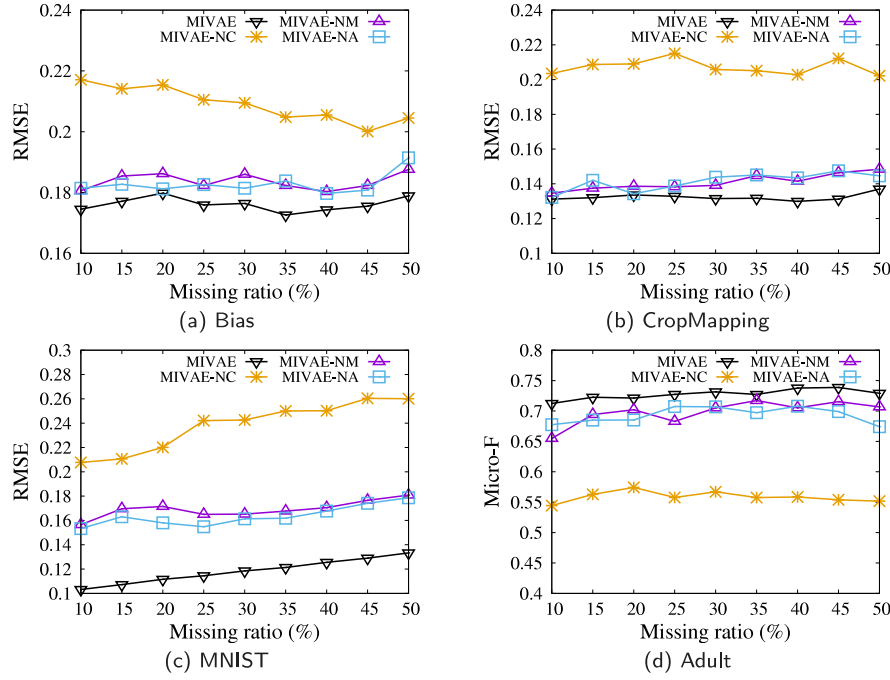


Fig. 7. Performance of each part of modification.

MICE and MIWAE over the MNIST. One possible reason is that the computation cost of PMM increases quadratically with the number of attributes. The MNIST is high-dimensional (with 784 dimensions), while the remain three datasets are relatively low-dimensional (with 23, 174, and 12 dimensions respectively).

4.6. Ablation study

As illustrated earlier, there are three modifications in the proposed MIVAE compared to the VAE model. To evaluate the necessity and effectiveness of each modification, we implement the MV imputation based on MIVAE model by discarding one modification, including (1) MIVAE-NC, discarding the data corruption layer in MIVAE; (2) MIVAE-NM, adopting a single latent sample rather than multiple samples for imputation; (3) MIVAE-NA, discarding the analysis layer and obtaining the final imputation results by averaging multiple candidate estimations.

Fig. 7 presents the performance of various algorithms over four real-world datasets. As shown, the imputation accuracies of approaches without one aspect of modification are worse than that of the proposed MIVAE, which verifies that each aspect of modification improves the imputation accuracy. Specifically, the gap between MIVAE and MIVAE-NC is the biggest, indicating that the data corruption layer contributes the most improvement for imputation accuracy. Through the data corruption layer, the synthetic MVs (noises) having the consistent missing pattern with the data to be imputed are introduced, making the model is more robust for MV imputation. By contrast, the imputation accuracies of MIVAE-NM and MIVAE-NA are close to each other, i.e., the gaps between them and MIVAE are almost the same, but smaller than MIVAE-NC.

4.7. Application in classification

In this subsection, we consider the application of data classification over the datasets of CropMapping, MNIST and Adult to further validate the effectiveness of our MIVAE. First, Fig. 8 shows some original complete digits adopted randomly from the testing set. Meanwhile, the corresponding incomplete and imputed (by MIVAE) digits are also

shown in Fig. 8. We replace the missing pixels with zeros for the incomplete digits, which means the incomplete digits also can be regarded as the imputed digits by employing zero imputation. As shown, the imputed and complete digits are similar, suggesting the incomplete digits are recovered well by applying MIVAE. Next, Fig. 9 reports the classification accuracies over the original complete data (denoted by *complete*), the data without imputation (denoted by *missing*), the imputed data by our MIVAE, existing MIDIA, and VAE. Notably, we adopt the MIDIA and SIVAE as the existing representative methods, because SIVAE is proposed based on existing VAE and MIDIA has the best performance in existing methods in most cases. Since the classification accuracy is generally related to the imputation accuracy, it is not surprising that the highest and lowest classification accuracies are achieved for original complete data and incomplete data without imputation, respectively. As shown in Fig. 9, MIVAE outperforms both MIDIA and SIVAE. Over the imputed data, the performance of SIVAE is the worst. Even when the missing ratio is greater than 30%, the performance of SIVAE is worse than that of the zero imputation, which verifies again that the performance of MV imputation by leveraging VAE directly is unsatisfactory. Moreover, the classification accuracy over the data set imputed by MIVAE is close to that of the original complete data, which further demonstrates the superiority of our MIVAE.

5. Conclusion and future work

In this paper, we propose a new generative model by exploring VAE, namely MIVAE, to support high-quality MV imputation. By sampling multiple latent samples from the posterior distribution of latent variables, and adding two additional layers, i.e., the data corruption layer and the data analysis layer, MIVAE is capable of effective multiple imputation. Experimental results over four real-world datasets demonstrate that MIVAE achieves higher imputation accuracy compared with existing MV imputation methods.

In general, the proposed MIVAE is valid under the assumption that the data are missing-at-random (MAR), which means that the missingness mechanism is independent of the missing values. MCAR and MNAR are stronger assumptions than MAR, and it is worth making attempts to develop models to missing data problems where the missing



Fig. 8. The complete samples of digits from the MNIST testing set, the corresponding incomplete samples by marking off 50% pixels, and the imputed samples obtained by MIVAE, MIDIA and SIVAE respectively.

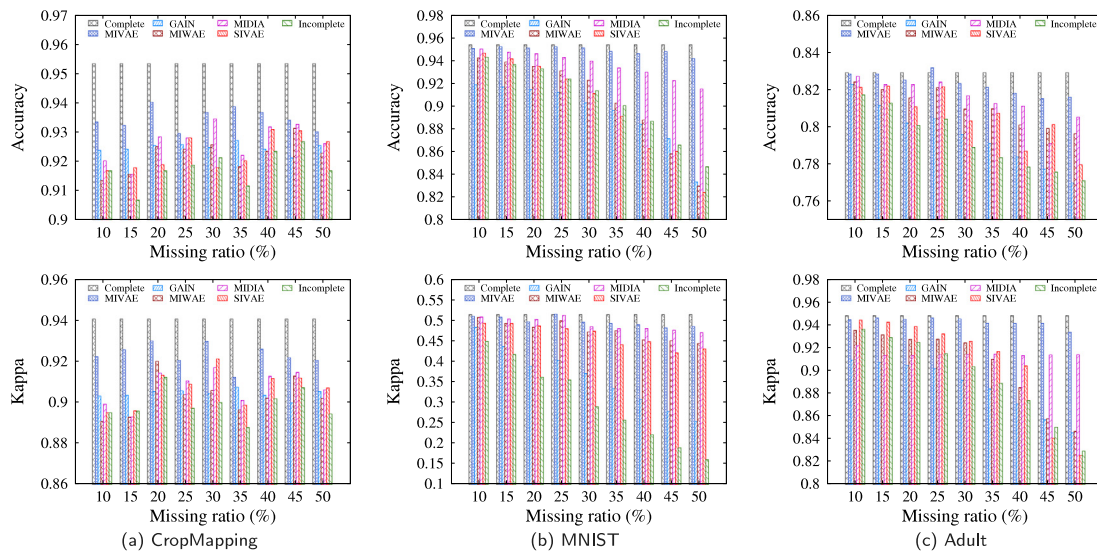


Fig. 9. Applications in classification.

mechanism is MCAR or MNAR in the future. On the other hand, most existing MV imputation approaches including the proposed MIVAE are designed for the tabular data. Designing effective models with suitable network architecture for data with complicated structure, e.g., graph data, is another interesting problem for future work.

CRedit authorship contribution statement

Qian Ma: Conceptualization, Methodology, Data curation, Writing – original draft. **Xia Li:** Conceptualization, Methodology, Software. **Mei Bai:** Methodology, Validation. **Xite Wang:** Writing – review & editing. **Bo Ning:** Supervision, Writing – review & editing. **Guanyu Li:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request

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